Chasena

Hot Powder Beverage Machine

IDE 402 x AI Project

Tools

Vizcom · Gemini · ChatGPT

Alena Tokel

Chasena is a countertop machine designed for **preparing powder-based beverages** such as matcha, hōjicha, and cacao, while maintaining **precision** and a **simple preparation ritual**.

User Insights

Users

- 24–32 year-old wellness-oriented urban users
- Drink matcha as a daily energy alternative to coffee
- Care about the ritual and aesthetic experience, relaible trust and consistent tasted
- Lack time and technical knowledge for traditional preparation

Pain Points

- Incorrect water temperature
- Inconsistent measurements
- Difficult whisking technique
- Multiple tools required for preparation
- Traditional tools are fragile or not always accessible
- Manual preparation can create mess, and bamboo tools are not dishwasher-saf

Key Insights

- The real challenge in matcha preparation is not time, but achieving precision and consistency without expertise.

Opportunity Areas

There is an opportunity to support precise preparation, reduce tool dependency, simplify the whisking process, and extend the system to other powder-based beverages while maintaining a clean and engaging preparation experience.

Problem Framing

‘’ How might we design a domestic matcha preparation machine that ensures precise temperature, dosing, and mixing while still allowing users to remain involved in the preparation process? ‘’

Inspiration

Concept

A precision-driven appliance that preserves the ritual of powder based drink preparation while simplifying the process for everyday users.

- Minimal
- Soft Geometric
- Clean
- Compact
- Precise

Sketches

Chasena
MAN

100 g
80 °C

Chasena

MATCHA
HOJICHA
Chasena
10 gr
Chasena

80
50
MAX
MIN
Chasena

100 g
80 °C
MAX
MAX

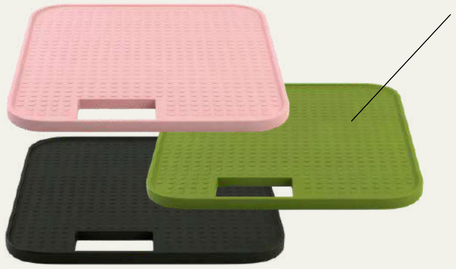
Chasena
10 gr
Chasena

0.0

80 °C
MAX

Components

Silicone rubber (long-lasting)



Food-grade stainless steel

Heat-resistant reinforced plastic

Stainless steel (recyclable)

Recycled aluminum & ABS plastic + matte finish

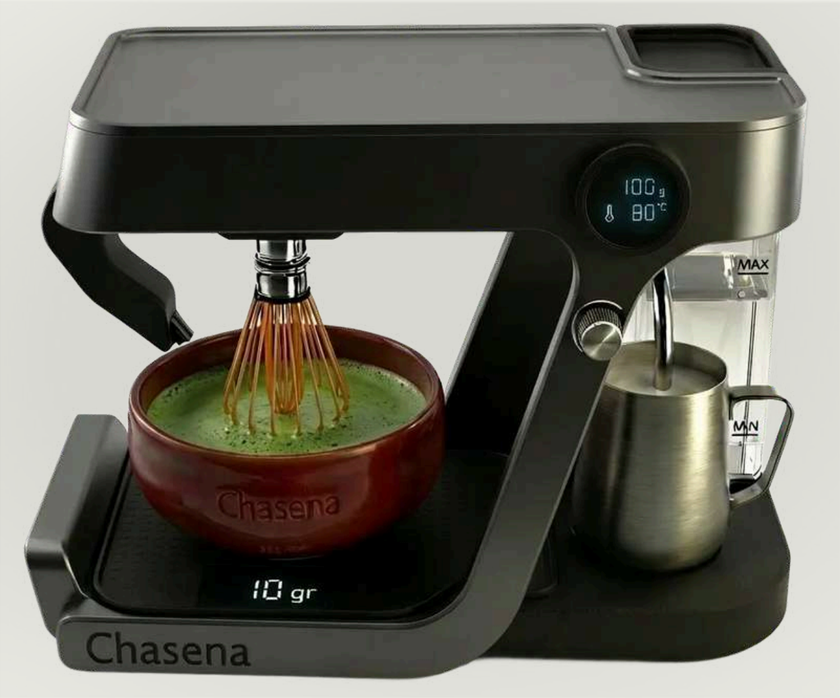
BPA-free Tritan plastic

Scale – Waterproof digital scale

Color Variants

A minimal palette inspired by Japanese ceramics, tea rituals, and natural materials.

Sakura Pink



Sumi Black

Kinako Cream

Impactful Prompts

- “Transform this sketch into a realistic product visualization while preserving the proportions and components shown in the drawing.”
- “Generate alternative product forms based on this concept direction while keeping the main functional components.”
- “Does this machine look feasible? What components might be missing in this design?”
- “How do beverage machines manage water flow, steam systems, and mixing mechanisms?”
- “Which whisk mechanism would work best for a powder-based beverage preparation machine with different bowl sizes?”

AI Role Tags

Clarifier

AI helped fill knowledge gaps during the research phase by explaining preparation mechanisms and measurement systems used in beverage machines.

Expander

AI supported the exploration of existing products and alternative preparation systems.

Re-framer

AI helped shift the concept from a single matcha device to a broader powder-based beverage preparation system.

Critique

AI reviewed sketches and helped identify missing components and usability issues.

Visualizer

AI tools such as Vizcom translated sketches into visual concepts to evaluate form, materials, and component relationships.

Simulation / Testing

AI animation tools were used to test component movement and reveal missing elements in the mechanism.

AI Response Summary

AI provided research insights about powder-based beverage preparation, including user behaviors, preparation challenges, and technical parameters such as temperature and powder ratios. It also explained mechanisms used in beverage machines and supported concept exploration by generating visual interpretations of sketches.

Reflection

Working with AI significantly accelerated the design process. Instead of spending excessive time on technical research and engineering explanations, AI quickly provided the necessary information and allowed the project to move forward faster. It also enabled rapid visualization of sketches, making it possible to explore multiple design directions and variations in a short time. This speed helped generate more sketches, experiment with different design languages, and evaluate forms through quick renders. Overall, AI supported a much faster exploration process while still allowing the designer to make the final decisions.

Design Decision

During the research phase, AI was used to rapidly explore information, mechanisms, and existing product systems related to hot beverage preparation. Relevant insights were accepted and integrated into the conceptual foundation of the project. However, AI-generated ideas were not adopted blindly. When suggestions did not align with the intended direction, they were rejected and redirected through new prompts and additional sketches. This iterative dialogue with AI enabled fast exploration of multiple design alternatives. Some concepts were transformed and refined, while others were intentionally discarded, allowing the project to evolve toward a clearer powder-based beverage preparation system.